

Text2SQL

Natural Language to SQL via
LLM-Modulo Framework

Group 27 · Project P13

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The Challenge

Language Gap

Natural language is ambiguous and context-dependent; SQL demands precise structure and explicit semantics.

Schema Complexity

Real-world databases feature multiple tables, intricate relationships, and constraints that require deep reasoning.

Security & Privacy

Improperly generated queries may expose restricted data or violate data governance policies.

Broad Accessibility

Non-technical users cannot easily formulate correct SQL — requiring dependence on technical experts.

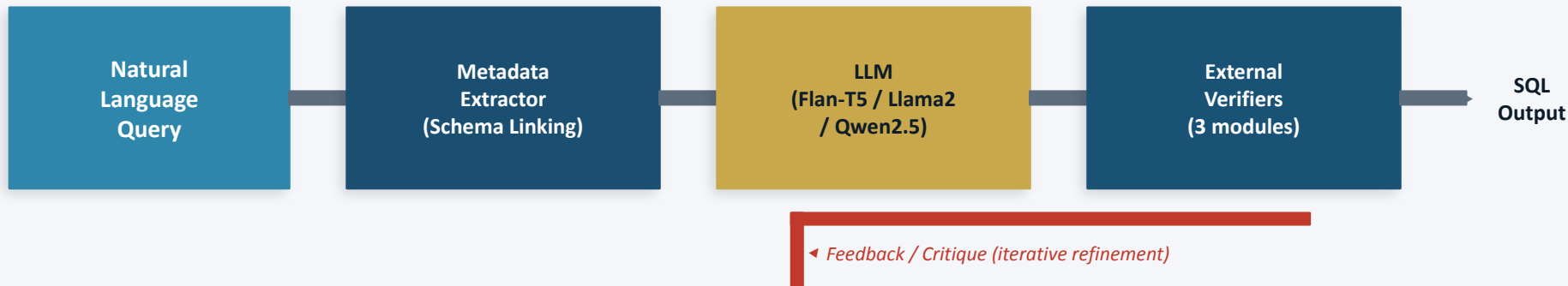
Goal

Enable intuitive, reliable, and safe database access for non-technical users through accurate Natural Language → SQL translation.

Key Research Questions

- Does targeted schema retrieval improve SQL generation accuracy?
- Can LLMs leverage external verifier feedback for iterative refinement?
- Which verifiers yield the highest accuracy gains per iteration?

LLM-Modulo Framework — Proposed Pipeline



External Verifiers

V1 Syntax Verifier	V2 Destructive Action & Auth Checker	V3 Constraint Verifier	V4 Query Graph Verifier
Evaluates SQL syntactic correctness and returns structural error feedback to the LLM for revision.	Blocks unsafe queries (DROP, DELETE) and enforces data access authorization constraints.	Performs dry-run execution to verify that all query constraints implied by the NL input are satisfied.	Converts each SELECT into a scoped subgraph, tracking context flow, dependencies, column bindings, and operator semantics to enable structured reasoning over nested SQL.

Baseline Methods

Bi-LSTM Seq2Seq

Traditional ML

Bidirectional LSTM architecture for NL→SQL (Finegan-Dollak et al.). Implemented via DyNet toolkit. Establishes classical supervised baseline.

Flan-T5

LLM Baseline

Instruction-finetuned encoder-decoder from Google. Evaluated in single-shot setting for Text2SQL generation.

Llama2

LLM Baseline

Open-source autoregressive LLM from Meta. Tested in both base and Text2SQL-finetuned variants.

Qwen2.5

LLM Baseline

Code-specialized LLM evaluated in single-shot setting; strong coding pretraining aids SQL generation.

State-of-the-Art Comparators

RESDSQL

Decouples schema linking from skeleton parsing using a ranking-enhanced encoder + skeleton-aware decoder.

DAIL-SQL

Optimizes in-context learning with Code Representation Prompts and DAIL selection for few-shot examples.

MAC-SQL

Multi-agent framework: Selector prunes schema, Decomposer applies CoT reasoning, Refiner corrects via execution.

CodeS

Open-source 1B–15B models; coarse-to-fine value retrieval (BM25 + LCS) for large schemas.

Spider Dataset

10,181

NL Questions

5,693

Unique SQL Queries

200

Databases

138

Domains

Key Characteristics

- Database-split evaluation — test DBs unseen during training
- Non-trivial foreign key relationships across tables
- Primary benchmark for ablation studies & schema retrieval analysis

BIRD Dataset

12,751

Text2SQL Pairs

95

Databases

33.4 GB

Total DB Size

37

Professional Domains

Key Characteristics

- Value grounding: reasoning over actual DB values, not just schema
- Noisy data, large tables, efficiency constraints — real-world realism
- Used to evaluate scalability of verifiers in enterprise-scale settings

Evaluation Metrics

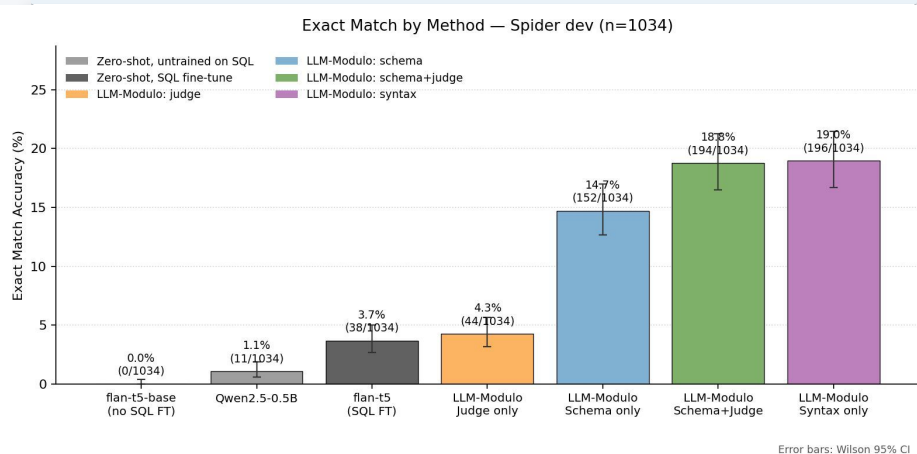
EX	EM	SVR	UQD
Execution Accuracy	Exact Match	Syntax Validity Rate	Unsafe Query Detection
Generated SQL produces the same result set as the ground-truth query when run against the target database.	Generated SQL exactly matches the reference query after normalization. Assesses structural correctness.	Percentage of generated queries that are syntactically valid and executable.	System's ability to identify and reject destructive or unauthorized SQL queries.

Ablation Study Design

- Full-schema prompt vs. schema-linked subset
- Verifier-free pipeline vs. incremental verifier addition
- Varying refinement iterations (1, 2, 3, ...) vs. accuracy

Method	Type	EX/EM
Bi-LSTM Seq2Seq	Classical ML	4.74%
Flan-T5 Base	LLM Baseline	0%
Qwen2.5-0.5B	LLM Baseline	1.06%
Flan-T5	LLM Baseline	3.68%
LLM-Modulo (Ours)	Proposed	24.53%

Exact Match by Method (Spider)



Key Observations

Schema Linking

Targeted metadata retrieval reduces irrelevant context, enabling the LLM to focus on query-relevant tables and relationships.

Verifier Impact

Syntax Verifier eliminates structural errors in early iterations; Constraint Verifier improves semantic accuracy in subsequent rounds.

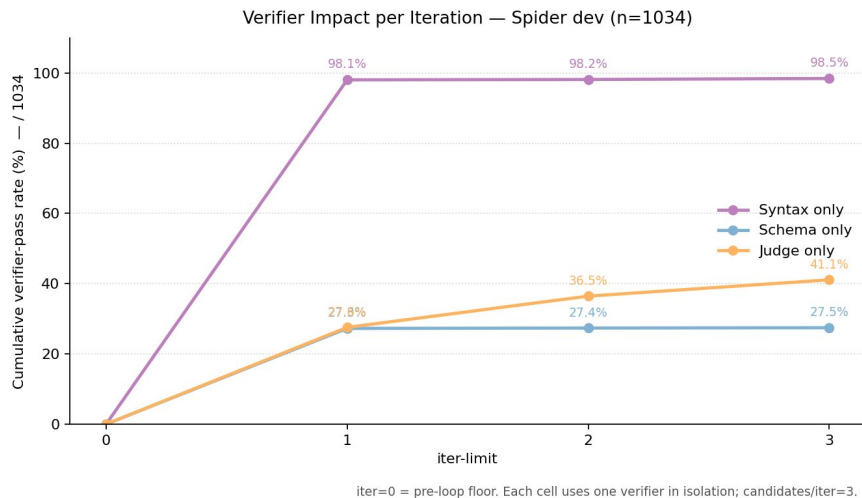
Safety

Destructive Action & Auth Checker successfully intercepts all DROP/DELETE test cases with no false positives observed.

Iteration Gains

Majority of accuracy improvement occurs within the first 1–2 refinement iterations, suggesting a cost-efficient stopping criterion.

Verifier Impact per Iteration



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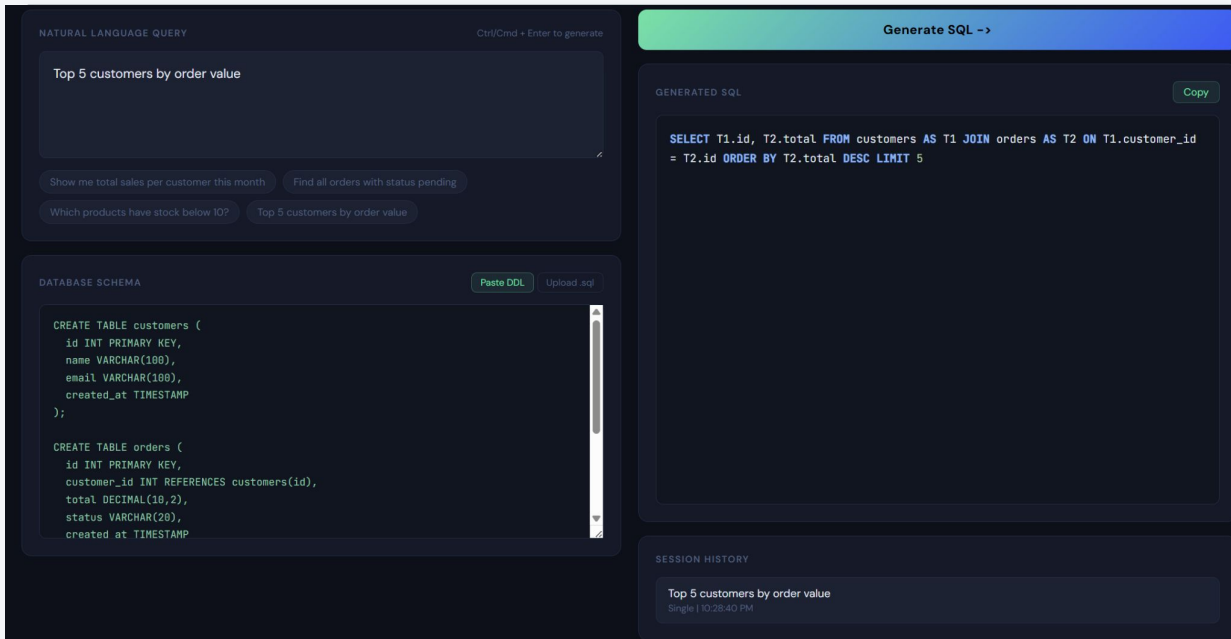
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UI Interface — Preliminary Design

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Design Highlights



Schema Insert

You can either paste DDL or upload a .sql file for database schema



Query History

Maintains session-level history of NL inputs and generated SQL Queries



SQL Syntax Highlighting

Code view with keyword highlighting for readability and manual review.

Project Plan — Tasks, Deadlines & Division of Work

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Timeline & Milestones

Deadline	Deliverable	Status
Mar 4	Dev environment, Flan-T5 setup, Metadata module	✓ Done
Mar 11	Dataset setup, LLM-Modulo skeleton, Frontend MVP	✓ Done
Mar 18	External verifiers + LLM baseline evaluation code	✓ Done
Mar 25	Verifier integration & full LLM-Modulo loop testing	✓ Done
Apr 8	Traditional ML baselines; proposed method results	✓ Done
Apr 15	Traditional ML results finalized	✓ Done
Apr 27	Final presentation completed	✓ Done
May 4	Final Report Submission	In Progress

Division of Work

Abhinav Gorantla

Dev env · Flan-T5 · Metadata · Verifiers · LLM-Modulo integration

Tejas Ajay Parse

LLM-Modulo core framework · Verifier pipeline integration

Rohitha Somuri

Dataset setup · LLM baseline evaluation · Experimental testing

Aryan B. Patel

Frontend interface design · Verifier integration with UI

Nandana Vikraman

Dataset setup · LLM baselines · Traditional ML models · Evaluation

Kamal Teja Annamdasu

Here is the Link for the repository for the project code

github.com/AbhinavGor/CSE573-Group27-Text2SQL



Questions?

Thank you for your attention
